

آزمون: بیوشیمی نظری مدت آزمون: 16 بر اساس دفترچه کاربر

بیوشیمی دیسپلینی

۱- از بتا اکسیداسیون کدام اسیدچرب، انرژی بیشتری آزاد می گردد؟

الف) اسید استئاریک ✓

ب) اسید لینولنیک

ج) اسید اولئیک

د) اسید لینولنیک

۲- کدامیک از عبارتهای زیر در مورد کنترل ترشح انسولین صحیح است؟

الف) با مصرف غذای چرب ترشح انسولین کاهش می یابد.

ب) در عفونت باکتریایی ترشح انسولین افزایش می یابد.

ج) در شرایط هیپوکسی و ورزش سنگین ترشح انسولین افزایش می یابد.

د) در شرایط استرس ترشح انسولین بواسطه اپی نفرین کاهش می یابد. ✓

منبع:

بیوشیمی لیبین کات

۳- کدام آنزیم در تبدیل باقیمانده VLDL به LDL نقش دارد؟

الف) لیپاز پانکراسی

ب) لیپوپروتئین لیپاز

ج) لیپاز حساس به هورمون

د) لیپاز کبدی ✓

۴- برای درمان فرد مبتلا به سرطان خون از داروی متوتروکسات استفاده می شود. اثر ضد سرطانی این دارو از طریق مهار سنتز کدامیک

از ترکیبات زیر است؟

الف) GMP

ب) CMP

ج) TMP ✓

د) AMP

منبع:

بیوشیمی لیبین کات

[link](#)

1. Beta-oxidation of which fatty acid releases more energy?

- a. Stearic acid ✓
- b. Linolenic acid
- c. Oleic acid
- d. Linoleic acid

2. Which of the following statements about the control of insulin secretion is correct?

- a. With consumption of fatty food, insulin secretion decreases.
- b. In bacterial infection, insulin secretion increases.
- c. Under conditions of hypoxia and heavy exercise, insulin secretion increases.
- d. Under stress conditions, insulin secretion decreases via epinephrine. ✓

Source: Lippincott Biochemistry

3. Which enzyme plays a role in converting VLDL remnant into LDL?

- a. Pancreatic lipase
- b. Lipoprotein lipase
- c. Hormone-sensitive lipase
- d. Hepatic lipase ✓

4. Methotrexate is used to treat a person with leukemia. The anti-cancer effect of this drug is through inhibition of the synthesis of which of the following compounds?

- a. GMP
- b. CMP
- c. TMP ✓
- d. AMP

5. A defect in all of the following factors leads to increased cholesterol, EXCEPT:

- a. Apo B100
- b. Apo E
- c. Apo B48 ✓
- d. LDL receptor

6. In Gilbert syndrome, serum unconjugated bilirubin and urine urobilinogen

- a. Decrease – Increase
- b. Decrease – Decrease
- c. Increase – Increase
- d. Increase – Decrease ✓

Source: Lippincott Biochemistry (Note: see comment at end — the "standard" expected pairing is increased unconjugated bilirubin with normal/decreased urine urobilinogen.)

7. Blood ALT level increases in which of the following conditions?

- a. Biliary duct obstruction – Hepatitis ✓
- b. Growing children – Skeletal muscle injury
- c. Growing children – Myocardial infarction
- d. Biliary duct obstruction – Bone disease *Source: Lippincott Biochemistry*

8. From the beta-oxidation of palmitic acid in the mitochondria, all of the following compounds are produced, EXCEPT:

- a. NADH
- b. NADPH+H⁺ ✓
- c. FADH₂
- d. Acetyl-CoA

9. All of the following statements about neonatal hyperbilirubinemia (jaundice) are correct, EXCEPT:

- a. The capacity for conjugation with glucuronic acid in the liver is low.
- b. Unconjugated bilirubin crosses the blood–brain barrier.
- c. Phototherapy makes unconjugated bilirubin more soluble.
- d. Uptake of unconjugated bilirubin by the liver is impaired. ✓

Source: Lippincott Biochemistry

10. For cholesterol synthesis, all of the following are required, EXCEPT:

- a. ATP
- b. NADPH+H⁺
- c. Malonyl-CoA ✓
- d. Acetyl-CoA

11. Which lipoprotein has the highest percentage of cholesterol?

- a. VLDL
- b. HDL
- c. Chylomicron
- d. LDL ✓

12. Which of the following statements is correct regarding the use of allopurinol (to reduce the clinical complications of gout)?

- a. Uric acid is converted to ammonia and carbon dioxide.
- b. It increases urinary excretion of uric acid.
- c. Uric acid is converted to allantoin.
- d. It increases urine xanthine levels. ✓

13. Which of the following statements about prolonged fasting is correct?

- a. Adipose tissue produces the ketone bodies needed by the body.
- b. Muscles use ketone bodies to produce energy. ✓
- c. Heart tissue uses glucose produced by gluconeogenesis to produce energy.
- d. The liver uses ketone bodies produced by the kidneys to produce energy. *Source: Lippincott Biochemistry*

14. In a person with the following lab results, which disease is likely? Increased ALP – Normal GGT

- a. Bone disease ✓
- b. Alcoholism
- c. Myocardial infarction
- d. Acute pancreatitis *Source: Lippincott Biochemistry*

15. Which one causes inhibition of the enzyme acetyl-CoA carboxylase?

- a. cAMP ✓
- b. Insulin
- c. Citrate
- d. Phosphatase

16. All of the following statements about heme synthesis are correct, EXCEPT:

- a. ALAS2 is controlled by the intracellular iron level.
- b. ALAS1 & ALAS2 are encoded by separate genes.
- c. Hemin decreases the expression and activity of ALAS1.
- d. ALAS2 is expressed in all tissues. ✓

17. Which of the following statements about pyrimidine metabolism is correct?

- a. Methylcytosine is converted to β -alanine.
- b. Uracil produces β -alanine. ✓
- c. Cytosine produces β -aminoisobutyrate.
- d. The pyrimidine ring is excreted from the body intact. *Source: Lippincott Biochemistry*

18. Which one plays a role in transporting free cholesterol from the plasma membrane to Apo A1?

- a. CETP
- b. LCAT
- c. ABCA1 ✓
- d. ACC

19. All of the following statements about deficiency of the enzyme hypoxanthine–guanine phosphoribosyltransferase (HGPRT) are correct, EXCEPT:

- a. It causes gout.
- b. Symptoms worsen with ethanol consumption.
- c. Uric acid production decreases. ✓
- d. Symptoms worsen with the use of thiazide diuretic drugs (for lowering blood pressure). *Source: Lippincott Biochemistry*

20. Which of the following statements about liver enzymes is correct?

- a. AST remains elevated in the blood for a longer time than ALT.
- b. The tissue distribution of ALT is specific to liver tissue. ✓
- c. In moderate liver injury, the AST/ALT ratio is greater than one.
- d. In chronic liver diseases, ALT level rises more than AST.

21. What is the final product of VLDL metabolism?

- a. LDL ✓
- b. HDL
- c. IDL
- d. Chylomicron